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Recent developments in electrochemical sensors for the detection and real-time monitoring of non-steroidal anti-inflammatory drugs

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ABSTRACT

The most commonly used medications to control fever, pain, and inflammation are non-steroidal anti-inflammatory drugs (NSAIDs). However, excessive use of them can have negative health consequences on humans, and their uncontrolled disposal in ecosystems raises serious environmental issues. Electrochemical sensors offer enormous potential for the sensitive, selective, effective, and economical detection of pharmaceuticals in complicated media, surpassing the drawbacks of conventional detection techniques. An overview of the most recent advancements in electrochemical sensors that use nanoscale materials as electrode modifiers that target NSAIDs, such as metallic nanomaterials, carbon-based materials, and hybrid materials, is provided in this review. Analysis of real samples, sensor/analyte interactions, and various electrode fabrication techniques was particularly taken into consideration. The studies in the literature were summarized in detail in terms of various analytical properties, such as linearity range, limit of detection (LOD), limit of quantification (LOQ), and detection method, among others. The experimental results were then discussed. Finally, this review discusses the potential, challenges, and opportunities in designing next-generation advanced sensing devices.

1. Introduction

In the earliest times, people preferred herbs and herbal medicines to treat themselves. According to the information in the Ebers papyrus, the first salicylate derivative was used for rheumatic pain 3500 years ago. After the advancement of technology and science and the discovery of acetylsalicylic acid by Felix Hoffman from Bayer's Industry, Germany, in 1879, non-steroidal anti-inflammatory drugs (NSAIDs) have become widely used [1,2]. NSAIDs are analgesic and anti-inflammatory drugs used to treat pain, fever, and inflammation in humans and animals. In addition, NSAIDs have not only analgesic and anti-inflammatory effects but also anti-cancer effects due to their antithrombotic effects. Since these drugs are available over the counter, they are the most widely used class of drugs worldwide [3]. Despite the wide therapeutic benefits of NSAIDs, they also have serious side effects. These side effects are cardiovascular, gastrointestinal toxicity, hypertension, kidney toxicity, and hepatotoxicity. Moreover, excessive use of NSAIDs can cause problems such as stomach ulcers and bleeding after injury [4]. Considering the side effects of these drugs due to long-term and excessive use, real-time monitoring of NSAIDs can allow for faster and easier early diagnosis of the side effects they cause.

According to recent studies in the European Union, NSAIDs (acetaminophen, ibuprofen, diclofenac, and naproxen) are the most commonly used drug group in aquatic environments. Therefore, when used excessively, they accumulate as residues in soil, wastewater, and environmental waters [4]. In fact, not only due to their use in aquatic environments but also due to improper disposal of unused drugs, residues can accumulate in the environment. This situation causes vital dangers to human and animal health, and environmental pollution. For this reason, the ability to analyze NSAIDs in environmental and biological environments is of great importance for the health of living beings and the protection of ecological balance. The most commonly used method for NSAID analysis in different real samples is the analytical methods [3,4].

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When the literature is examined, traditional analytical techniques (high-performance liquid chromatography (HPLC), ultraviolet-visible spectrometry (UV–VIS), and gas chromatography (GC)) have been used for the analysis of NSAIDs [1,4–7]. These analytical techniques are robust and sensitive, but they have several disadvantages, including sample preparation, lengthy analysis times, expensive and complex equipment, and high chemical usage. Due to these disadvantages, it becomes challenging to meet the increasing needs of modern pharmaceutical industries in the future. However, electrochemical methods stand out as alternative analytical methods due to their affordability, environmental friendliness, minimal use of chemicals, simplicity, and reliability [4,8,9]. Moreover, the development of different types of sensors using electrochemical methods is becoming an active and robust research in drug analysis. Electrochemical sensors offer excellent sensitivity, selectivity, user-friendliness, and portability, and can be integrated into miniature devices and in situ analysis in drug analysis. These sensors are of great interest in the rapid and sensitive detection of NSAIDs [10,11]. Additionally, electrochemical sensors offer cost-effective analysis with minimal operational and maintenance requirements, require simpler setups, and reduced training demands, allowing non-specialists to perform analyses without extensive technical expertise. They also offer potential for portability and point-of-care use, as well as lower logistical costs associated with sample transport and centralized testing.

Although several review articles have addressed electrochemical detection of pharmaceutical pollutants or NSAIDs in general, no comprehensive review to date has specifically focused on the development and application of hybrid nanomaterial-based electrochemical sensors for NSAID detection, particularly in complex real-world matrices such as biological fluids and environmental samples. This paper fills that critical gap by systematically reviewing and comparing the performance of these hybrid nanostructures, discussing their synergistic properties, and highlighting design strategies that improve selectivity, sensitivity, and operational stability. Furthermore, this review offers original insights into sensor scalability and real-time monitoring capabilities, which are essential for practical field applications. The novelty of this work lies in its focused analysis of recent developments (2020–2024), its attention to matrix-specific challenges, and its emphasis on the practical translation of lab-scale sensors into real-world diagnostic and environmental tools.

This review provides an overview of electrochemical sensors developed for the rapid and sensitive analysis of NSAIDs since 2020. The main purpose of this review is to examine in detail the current developments of NSAID drugs, their widespread use, new methods developed in environmental monitoring, and the most recent studies. In addition, the principles, sensitivities, properties, and strategies of nanomaterials and their effects on electrochemical sensors are explained in detail in this review. The basics, mechanisms, and future prospects of sensors used in the analysis of NSAIDs in their application areas will be discussed. For this purpose, NSAID drugs, the importance of electrochemical studies performed for these drugs, and their applications are examined and discussed in detail. Additionally, this review will provide insight into future electrochemical studies for the selective and sensitive analysis of NSAIDs.

2. Overview of drugs used in NSAIDs

2.1. Diclofenac

Diclofenac (DIC) is an NSAID widely prescribed to treat conditions such as osteoarthritis, rheumatoid arthritis, ankylosing spondylitis, and acute muscle pain. It is also combined with nucleoside metabolic inhibitors in specific cancer treatments. While generally safe at recommended doses, overdose can cause serious side effects, including cardiovascular risks, especially in patients recovering from coronary artery bypass surgery. Additionally, DIC has become a significant environmental pollutant, frequently detected in aquatic ecosystems, posing a toxic threat to organisms such as algae, zooplankton, and invertebrates. Its persistence in the environment, even in trace amounts, has highlighted the need for improved methods to monitor and remove it from water bodies [12,13].

2.2. Naproxen

Naproxen (NAP) is known for its effectiveness in treating pain, inflammation, and fever. It is commonly prescribed for conditions like arthritis, migraine, and musculoskeletal disorders. NAP can lead to adverse effects, particularly on kidney function and the gastrointestinal system, especially when misused or overdosed. Its metabolites also pose environmental concerns, as they can accumulate in water sources and affect aquatic life. Therefore, accurate and reliable methods to measure NAP levels in biological and environmental samples are essential. These include electrochemical techniques, which are gaining popularity due to their affordability, sensitivity, and ease of use for detecting NAP in various settings [14–16].

2.3. Mefenamic acid

Mefenamic acid (MFA) is commonly used to manage pain and inflammation, particularly in conditions like osteoarthritis, rheumatoid arthritis, and menstrual cramps. It works by inhibiting the production of prostaglandins, which are responsible for triggering pain and inflammation. Despite its therapeutic benefits, prolonged or excessive use of MFA can lead to serious side effects, including gastrointestinal issues, kidney problems, and cardiovascular risks. Additionally, MFA residues in the environment, especially in water bodies, can threaten aquatic life, affecting reproductive health and behavior. Therefore, accurate detection and monitoring of MFA in biological and environmental samples are crucial to ensure safety and prevent ecological damage [17–19].

2.4. Ibuprofen

Ibuprofen (IBU) is the drug of choice for relieving pain, reducing fever, and treating conditions such as headache, muscle pain, and inflammatory disorders. It is often sold over the counter, which can lead to accidental or intentional misuse, which can lead to significant health risks, including gastrointestinal problems, heart problems, and an increased risk of heart attack. In addition to its health effects, IBU is an important environmental pollutant commonly found in wastewater and can have adverse effects on aquatic organisms, including fish embryos and microorganisms. Given its widespread use and environmental presence, accurate and accessible methods for detecting IBU in biological and environmental samples will be necessary for safety and ecological protection [20–22].

2.5. Aspirin

Aspirin (ASP) is a drug known for its analgesic and antipyretic properties, which are used to treat a variety of conditions, including cardiovascular disease, cancer, and Alzheimer's. It also has antioxidant effects that help neutralize harmful free radicals in the body. However, the metabolism of ASP produces salicylic acid, which can be toxic at high levels and lead to complications such as ulcers and stomach damage. Beyond its medical applications, ASP has become a common environmental pollutant, often detected in wastewater and water bodies. Due to its persistence in aquatic environments, ASP can pose risks to aquatic organisms, such as free radical production and DNA damage. As a result, improved detection methods are needed to reduce its harmful environmental effects and prevent ecological damage [23–25].

2.6. Nimesulide

Nimesulide (NIM) is an NSAID with analgesic, antipyretic, and anti-inflammatory properties. It is commonly used to treat acute pain, osteoarthritis, and dysmenorrhea by inhibiting cyclooxygenase-2 (COX-2), an enzyme involved in the processes of pain and inflammation. While it is effective and widely used, NIM carries significant risks, particularly when used in high doses or over extended periods. Serious side effects include hepatotoxicity, leading to liver failure, the need for transplantation, as well as kidney damage. Due to these risks, NIM has been banned in some countries like the UK and Germany [26]. Furthermore, its persistence in water makes it an emerging environmental pollutant, posing potential risks to both ecosystems and human health [27,28].

2.7. Indomethacin

Indomethacin (IDM) is preferred due to its analgesic, antipyretic, and anti-inflammatory effects. It is known to be especially effective in treating chronic musculoskeletal pain, osteoarthritis, rheumatoid arthritis, and gout. IDM works by inhibiting cyclooxygenase enzymes (COX-1 and COX-2), which are responsible for producing prostaglandins that mediate pain, inflammation, and fever. While it is beneficial for managing these conditions, IDM is associated with side effects, including gastrointestinal issues and kidney damage, especially when used long-term [29,30].

2.8. Piroxicam

Piroxicam (PRX) is used to treat rheumatoid arthritis, osteoarthritis, and other inflammatory diseases and is known for its analgesic and antipyretic properties. PRX works by inhibiting the production of prostaglandins, thereby reducing pain and inflammation. However, its long-term or excessive use can lead to severe side effects, including kidney failure, liver damage, and gastrointestinal issues like ulcers and bleeding. Additionally, PRX is found in wastewater and aquatic environments, where it poses a risk to ecosystems even at low concentrations. Given its widespread use and environmental impact, accurately determining PRX levels is critical to protecting human health and natural habitats [31–33].

2.9. Ketoprofen

Ketoprofen (KTP) is commonly preferred for the treatment of pain, particularly from musculoskeletal disorders and inflammatory diseases, due to its analgesic and anti-inflammatory properties. However, its widespread use and improper disposal, combined with inefficiencies in wastewater treatment, lead to its persistence in the environment. KTP, in particular, is a persistent contaminant that is not easily degraded by conventional treatment methods and is widely present in wastewater effluents, surface waters, and even drinking water [34,35]. Indications, side effects, environmental concerns, metabolites, and detection approaches for NSAIDs are summarized in Table 1.

3. Classification of nanomaterials used in the analysis of NSAIDs

In recent years, nanomaterials have been widely used in the design of electrochemical sensors. These nanomaterials exhibit excellent chemical, physical, and electrochemical advantages in sensor design. Moreover, nanomaterials contribute to the sensor's performance, stability, selectivity, sensitivity, and polymeric structure formation. Nanomaterials used in sensor design offer many advantages. Carbon-based nanomaterials and metallic nanomaterials, in particular, are widely used in electrochemical sensing. Carbon-based nanomaterials offer many significant advantages due to their high chemical and thermal stability, conductivity, tunable structure, texture, and morphology, low cost, and easy availability. Metallic nanoparticles, which are zero-dimensional nanoscale metals ranging from 1 to 100 nm, possess properties such as controllability of shape, composition, size, structure, assembly, and optical properties. This section provides detailed information about the nanomaterials most commonly used in sensors developed with NSAID drugs.

Table 1
Summary of major NSAIDs: indications, adverse effects, environmental concerns, metabolites, and detection approaches.

NSAID	Medical Indications / Beneficial Effects	Adverse Effects / Toxicity	Environmental Persistence & Impact	Metabolites	Relevant Sample Types & Common Detection Methods	Ref.
DIC	Osteoarthritis, rheumatoid arthritis (RA), ankylosing spondylitis, acute pain; sometimes cancer therapy	Cardiovascular risks, gastrointestinal (GI) bleeding	Aquatic pollutant; toxic to algae, zooplankton, invertebrates	Hydroxylated & conjugated metabolites	Serum, urine, wastewater; HPLC, GC, electrochemical sensors	[12,13]
NAP	Pain, arthritis, migraine, fever	GI bleeding, renal toxicity	Accumulates in water sources; affects aquatic life	O-desmethyl naproxen	Plasma, urine, surface water; LC, UV–VIS, electrochemical methods	[14–16]
MFA	Menstrual pain, osteoarthritis, RA	GI ulceration, kidney problems, Cardiovascular risks	Wastewater pollutant; affects reproduction in aquatic organisms	Hydroxylated derivatives	Urine, blood, environmental waters; HPLC, electrochemical sensors	[17–19]
IBU	Pain, fever, headache, inflammation	GI issues, Cardiovascular risks, nephrotoxicity	Common wastewater pollutant; harmful to fish embryos, microbes	Hydroxy- and carboxy-ibuprofen	Serum, urine, river water; HPLC, GC–MS, electrochemical techniques	[20–22]
ASP	Analgesic, antipyretic; prevention of CVD, cancer, Alzheimer's	Ulcer risk, gastric bleeding, salicylic acid toxicity	Persistent in aquatic environments; causes oxidative stress	Salicylic acid	Plasma, urine, wastewater; spectrophotometry, LC, electrochemical sensors	[23–25]
NIM	Acute pain, osteoarthritis, dysmenorrhea (COX-2 selective)	Severe hepatotoxicity, renal injury (banned in some countries)	Persistent in surface waters; ecological risk	NR	Serum, urine, river water; LC-MS, electrochemical assays	[26–28]
IDM	RA, osteoarthritis, gout, musculoskeletal pain	NR	Detected in wastewater (less studied)	NR	Plasma, formulations; HPLC, voltammetry	[29,30]
PRX	RA, osteoarthritis, inflammatory diseases	GI bleeding, renal & hepatic toxicity	Stable in wastewater; ecotoxic even at low conc.	Hydroxylated conjugates	Plasma, urine, surface waters; LC, electrochemical sensors	[31–33]
KTP	Pain from musculoskeletal disorders, inflammation	GI ulceration, renal issues	Persistent in effluents; detected in drinking water	NR	Wastewater, fish tissues, plasma; LC-MS, electrochemical detection	[34,35]

NR: Not reported, RA: Rheumatoid Arthritis, CVD: Cardiovascular Disease, GI: Gastrointestinal, HPLC: High Performance Liquid Chromatography, GC: Gas Chromatography, GC–MS: Gas Chromatography–Mass Spectrometry, LC: Liquid Chromatography, LC-MS: Liquid Chromatography–Mass Spectrometry, UV–VIS: Ultraviolet–Visible Spectroscopy, COX-2: Cyclooxygenase-2, MS: Mass Spectrometry.

3.1. Metallic nanomaterials

Metallic nanomaterials play an important role in the electrochemical detection of NSAIDs due to their outstanding electrical conductivity, fast electron transfer, excellent physical and chemical stability, excellent selectivity, corrosion resistance, and surface tunability [36,37]. Due to these advantages, these nanomaterials are widely used in biological, environmental, and food fields. Commonly used metallic nanomaterials include gold (Au), silver (Ag), platinum (Pt), and copper (Cu), all of which contribute to enhanced sensor performance [36]. Gold nanoparticles (AuNPs) offer significant advantages in the design of electrochemical sensors, including a large surface area, stability, optical properties, and the ability to analyze at very low concentrations [36,37]. In addition, when the literature is reviewed, AuNPs have been widely used in the analysis of the active ingredients of NSAID drugs NAP [38] and DIC [39]. Upon reviewing the studies, it has been demonstrated that AuNPs significantly enhance the electron transfer rate, resulting in increased sensitivity and a lower limit of detection (LOD) for NSAID analysis [36]. Silver-based nanomaterials, including silver nanoparticles (AgNPs) and silver nanowires (AgNWs), offer greater electrical conductivity than AuNPs and are more cost-effective, making them attractive for large-scale sensor development [37]. AgNWs have been used in the design of both modified and hybrid sensors for NSAID detection in pharmaceutical formulations [40]. These nanoparticles facilitate real-time, label-free detection of NSAIDs while maintaining excellent mechanical flexibility, making them suitable for wearable or portable electrochemical sensing applications [36]. Copper nanoparticles (CuNPs) are gaining attention due to their high availability, affordability, and excellent redox properties. Copper-modified electrodes enhance the detection of NSAIDs in complex biological fluids and environmental samples, facilitating widespread applications in pharmaceutical quality control and environmental monitoring [36,41]. Platinum-based nanomaterials (PtNPs) can be integrated with other nanomaterials in electrochemical sensors, and these nanoparticles exhibit superior electrocatalytic activity and chemical stability, making them effective for highly sensitive electrochemical detection of NSAIDs [36,42,43].

Among the NSAIDs reviewed, AuNPs were most frequently applied in the detection of NAP [38] and DIC [39], providing significant improvements in sensitivity and LOD. Silver [40] and copper nanoparticles [44] were more commonly used in pharmaceutical formulations and biological fluids for IBU detection due to their conductivity and cost-effectiveness. Comparatively, AuNPs showed the highest enhancement in analytical performance for NAP [38], while CuNPs proved more effective in biological matrices for IBU [44].

3.2. Carbon-Based nanomaterials

Carbon-based materials have garnered significant attention from researchers in flexible sensing due to their remarkable properties, such as lightweight, high strength, superior electrical conductivity, excellent chemical and thermal stability, exceptional flexibility, and adaptability. Due to these important properties, carbon-based nanomaterials are widely used in electrochemical studies. When the literature is examined, carbon nanotubes (CNTs), graphene (GR), graphene oxide (GO), carbon nanofibers (CNFs), and other carbon-based nanoparticles are widely used in sensor design [36,37]. Materials like carbon nanotubes (CNTs) and graphene stand out for their high surface area and exceptional conductivity [36]. Functionalized graphene oxide (GO) has been utilized to achieve high specificity in detection, while multi-walled carbon nanotubes (MWCNTs) have been effectively employed in the fabrication of sensors, offering enhanced responsiveness for NSAIDs detection [45].

3.2.1. Carbon nanotubes

CNTs are nanomaterials in the form of graphite sheets wrapped in a cylinder with a diameter of <100 nm [46]. These nanomaterials have excellent chemical and physical properties. In addition, CNTs are divided into groups as single-walled carbon nanotubes (SWCNTs) and multi-walled carbon nanotubes (MWCNTs) according to the amount of carbon they contain. MWCNTs, which have two or more carbon numbers, are nanomaterials with diameters ranging from 1 to 3 nm. SWCNTs, on the other hand, are structures with a single carbon number and are nanomaterials with a diameter ranging from 0.4 nm to 2 nm. CNTs are widely used in the design of electrochemical sensors due to their advantages, such as exceptional stability, fast electron transfer, excellent electrical properties, and sensitivity. These nanomaterials can positively affect the performance of the electrodes used in surface modifications, electrode attachment methods, and the presence of electron mediators [37,46].

MWCNTs have been predominantly used for the detection of ASP [47] and PRX [48], offering rapid electron transfer and improved response signals. SWCNTs, however, are less widely used than MWCNTs. Overall, MWCNTs provided more versatile application across different NSAIDs and sample types, particularly in pharmaceutical and serum-based matrices.

3.2.2. Graphene and graphene derivatives

Discovered in 2004, GR is a two-dimensional, single-layered nanomaterial composed of atoms formed by sp2 bonds [49,50]. GRs offer several important advantages, including mechanical flexibility, chemical and physical stability, biocompatibility, rapid electron transfer, and a large surface area. Moreover, GR and its derivatives have attracted great attention in the design of electrochemical sensors in recent years due to their thermal, optical, and electrical properties [51]. GO and reduced graphene oxide (rGO) are graphene derivatives. GO offers advantages due to its excellent thermal and electrical conductivity, large surface area, easy modification, and versatile structure. They are distinguished from GR due to the –OH, –COOH, –CO, and epoxy groups found in GO [37,46]. Therefore, GOs have a more stable structure, but their conductivity is lower than GRs [37,46]. rGO has advantages, including rich properties in terms of electron transfer and bipolar load capacity. Due to these significant advantages and wide application areas, GR and its derivatives have attracted great attention in the design of electrochemical sensors for selective and sensitive analysis of NSAIDs.

Graphene-based nanomaterials, especially GO and rGO, have shown superior performance in the detection of NAP [52,53] and DIC. rGO, with its higher conductivity, achieved lower LOD values in several studies, particularly in complex matrices such as wastewater and blood serum [54,55]. When comparing GO and rGO, rGO consistently offered enhanced sensitivity, especially in electrochemical setups aimed at NAP quantification [52,53].

3.3. Hybrid materials

Hybrid materials have emerged as a transformative approach in electrochemical sensors for NSAID detection, combining the advantages of organic and inorganic components to enhance stability, sensitivity, and selectivity. For instance, hydrogel-based hybrids, such as calcium alginate/calcium silicate (CaAlg/$CaSiO_3$), mitigate swelling issues in molecularly imprinted polymers (MIPs), thereby preserving cavity integrity for precise biomolecule recognition. This principle can be adapted to NSAID sensing. Similarly, conductive hybrids, such as MWCNT-coated microcilia/porous PDMS structures, demonstrate how synergistic effects amplify sensitivity and broaden detection ranges, which is crucial for real-time NSAID monitoring in complex matrices. Innovations such as MOF-based hybrids address stability and catalytic activity limitations, enabling both enzymatic and non-enzymatic detection mechanisms. Additionally, 3D-printed flexible sensors incorporating graphene-PEDOT: PSS composites highlight the potential of hybrid systems for multifunctional sensing, including pharmaceutical analytes. By integrating advanced materials (e.g., nanoparticles, polymers, and carbon-based fillers), hybrid sensors achieve improved signal-

to-noise ratios, durability, and adaptability—key features for wearable or point-of-care NSAID monitoring. Future efforts should focus on optimizing hybrid interfaces and scalable fabrication to advance real-world applications [56–60].

Hybrid nanomaterials have been applied across a broad spectrum of NSAIDs, including IBU, IDM, and DIC. In particular, MWCNT- or PEDOT-based hybrids were frequently reported to enhance selectivity and extend the linear range. Comparatively, hybrid systems performed best for IDM in complex matrices such as plasma, due to their matrix compatibility and antifouling properties [45].

4. Applications of electrochemical sensors designed for the detection of NSAIDs

Electroactive nature of NSAIDs has recently attracted interest in electrochemical studies, particularly for applications in therapeutic monitoring, environmental analysis, and pharmaceutical quality control [17]. Advancements in material science have led to the development of modified electrode surfaces, incorporating nanocomposites and advanced materials, such as graphene oxide, to enhance the electrocatalytic oxidation and detection of NSAIDs. These innovations improve sensitivity and selectivity, enabling the detection of trace levels in pharmaceutical formulations and biological fluids. Furthermore, they play a crucial role in monitoring drug stability, assessing pharmacokinetics, and detecting residues in environmental matrices, contributing to both healthcare and environmental protection [61]. In summary, scientific advances have improved the electrochemical detection of NSAIDs, enabling sensitive and selective methods for therapeutic monitoring, pharmaceutical analysis, and environmental safety.

Electrochemical analysis is a simple and low-cost method for determining the concentrations of electroactive species in a solution, both qualitatively and quantitatively. Compared to techniques like chromatography and spectroscopy, electroanalytical methods offer several advantages, including ease of use, high sensitivity and reliability, short measurement times, and environmental friendliness. Researchers are actively developing advanced techniques and next-generation sensor designs to enhance the detection of NSAIDs. In recent years, nanoparticle-doped electrochemical sensors designed for the analysis of NSAIDs have been frequently used in various biological and pharmaceutical samples due to their advantages, such as rapid electron transfer, stability, sensitivity, and easy preparation.

In addition to sensor development, successful detection in complex biological and environmental matrices often requires appropriate sample pre-treatment strategies to reduce matrix effects and enhance compatibility. For biological samples such as serum or plasma, protein precipitation using organic solvents or acids is commonly employed to prevent electrode fouling. For instance, a protein removal step was applied prior to IBU detection in human serum using a Cu_3TeO_6/GCE sensor [44]. In environmental samples such as wastewater, SPE is frequently used to concentrate analytes and eliminate interferents, as demonstrated in several studies targeting DIC and NAP [54,78]. Dilution and filtration are also widely used for urine samples to reduce matrix complexity. Additionally, magnetic separation techniques and molecular recognition-based extraction strategies, such as those employing MIPs, have been broadly reported in the literature as effective tools for enhancing selectivity and sensitivity in pharmaceutical and biological sample preparation. These pre-treatment approaches are critical in improving the reliability, sensitivity, and applicability of electrochemical sensors for real-sample analysis [54,67].

The results of the studies published in the literature between 2020 and 2025 to detect NSAIDs were presented in detail (Table 2).

Kokulnathan et al. [62] developed a novel electrochemical platform for DIC detection using flower-like bismuth oxyhalides (BiOX, $X = Cl, Br, I$) synthesized via a solid-state mechanochemical method. The BiOI-modified GCE demonstrated superior electrochemical performance, characterized by its high conductivity, large surface area, and abundant electroactive sites. The electrochemical behavior of the BiOI/GCE was studied using DPV for diclofenac detection, with the sensor showing linearity in the concentration ranges of 0.5–11.3 μM and 11.3–76 μM. The LOD was determined to be 0.11 μM, with a 1.33 μA μM^{-1} cm^{-2} sensitivity. The sensor exhibited excellent selectivity, repeatability, and stability, with recovery studies in human urine and river water samples confirming its accuracy and reliability. The BiOI/GCE sensor's advanced electrocatalytic properties make it a promising candidate for clinical diagnostics and environmental analysis (Fig. 1**C).** Kimuam K. et al. [54] designed an electrode surface modified with platinum nanoflowers and reduced graphene oxide nanocomposite (PtNFs/rGO) using cyclic voltammetry (CV). Systematically examined using the CV method, they observed a five-fold increase in the analytical peak current of the PtNFs/rGO electrode compared to a bare electrode. The designed PtNFs/rGO electrode not only accelerated electron transfer with DIC but also promoted DIC accumulation on the electrode surface, providing electrochemical sensitivity for quantitative analysis of DIC. The calibration curve for quantitative DIC analysis using the DPV method was obtained linearly in the range of 0.1–100 μM, and the detection limit was found to be 40 nM ($S/N = 3$). In addition, the designed sensor was successfully applied for the determination of diclofenac in human urine samples (Fig. 1**A)**. Killedar L. et al. [63] developed a ruthenium-doped titanium dioxide (Ru-TiO_2) modified electrochemical sensor for the quantitative analysis of DIC. SEM, TEM, EDX, and XRD techniques were used to characterize Ru-TiO_2 nanoparticles. Electrochemical studies were performed using the SVW technique. The results showed a limit of detection (LOD) of 1.48 nM, lower than previously reported. The sensitivity and selectivity of the Ru-TiO_2-based sensor developed for the quantitative analysis of DIC were observed to be increased. The designed sensor is recommended for both quality assurance and the effective determination of DIC in biological and clinical samples (Fig. 1**B)**.

In another study by Mutharani et al. [44], a facile wet-chemical synthesis method was employed to create a novel 3D stone-like copper tellurate (Cu_3TeO_6) electrocatalyst. The Cu_3TeO_6/ GCE sensor was fabricated by modifying the electrode surface with the synthesized copper tellurate nanostructures. The electrochemical determination of IBU was performed using DPV. Under optimal conditions, the Cu_3TeO_6/GCE sensor demonstrated linear detection ranges of 0.02–5 μM and 9–246 μM, with a sensitivity of 5.97 μA μM^{-1} cm^{-2} and a low LOD of 0.017 μM. The accuracy and reliability of the sensor were validated through recovery studies using human urine and blood serum samples, yielding recovery rates of 96 % to 98.6 %. The sensor also exhibited high selectivity, reproducibility, and stability, highlighting its potential for practical applications in clinical and environmental analyses (Fig. 2**A).** Nair A. et al. [40] developed a novel molecularly imprinted (MIP)-based electrochemical sensor for the selective and sensitive analysis of IBU on acid-functionalized graphene oxide (Af-GO) decorated with silver nanoparticles (AgNPs). They used *Caesalpinia sappan* leaf extract for AgNP synthesis. The synthesized AgNPs effectively decorated the Af-GO onto which the MIP was grafted. The synthesized AgNPs were effectively decorated on the Af-GO onto which the MIP was grafted. The findings indicate that AgNPs@Af-GO-MIP has good selectivity and detection ability against IBU. The detection limit of AgNPs@Af-GO-MIP was found to be directly proportional to the IBU concentration, with a detection limit of 8.7×10^{-9} mol L^{-1} ($S/N = 3$). They proved the selectivity of the developed sensor by using interfering substances. Most importantly, the sensing capability of the developed AgNPs@Af-GO-MIP was efficiently demonstrated in pharmaceutical and human urine samples, which offers a glimpse into the real-time applicability of the sensor.

An innovative electrochemical sensor was developed by Feyziazar et al. [45] for the determination of IDM using a GCE modified with a novel chitosan (CS)-copper nanocubic (Cu NC) composite. The CS-Cu NC/GCE sensor was designed by electrochemically depositing Cu NCs on a copper electrode, followed by chitosan modification to enhance

Table 2
Applications of electrochemical sensors for the use of nanoparticle-based platforms in the analysis of biological and pharmaceutical samples.

ANALYTE	MODIFIED ELECTRODE	METHOD	LOD	LINEAR RANGE	REAL SAMPLE	RECOVERY (%)	REF.
DIC	Cu–Al LDH/CFYM/GPGE	DPV	1.91 μM	1.99 – 23.44 μM	Human blood serum Tablets	96.00 – 103.75 94.63 – 102.06	[69]
	COOH–CNTs/ZnO/NH_2–CNTs/GCE	SWASV	0.077 μM	0.021– 0.075 μM	Tablet Serum Urine Artificial Saliva/Sweat Drinking Water Tap Water	101.00 – 102.00 97.00 – 99.00 98.00 – 101.00 97.00 – 99.00 96.00 – 101.00 99.00 – 102.00	[70]
	IONPs/CPE	DPV	2.45 nM	0.01 – 100 μM	Blood Serum	95.30 – 106.00	[71]
	PtNFs/RGO/SPE	DPV	0.04 μM	0.1 – 100 μM	Human Urine	84.00 – 105.00	[54]
	Coreshell/GCE	SWV	0.0341 μM	0.01 – 300 μM	Pharmaceutical formulations Human urine	96.60 – 98.20 94.00 – 97.50	[72]
	Ru–TiO_2/CPE	SWV	0.00148 μM	0.1 – 2.0 μM	Tablet Human urine	93.00 – 99.20 97.30 – 98.75	[63]
	MIP/SPCE	DPV	70 nM	0.1 – 10 μM	Tap water River water	88.00 – 92.00 108.00 – 115.00	[73]
	CeO_2–SPCE	SWV	0.4 μM	0.4 – 26 μM	real water biological samples	NR	[74]
	PEDOT/GCE	DPV	9.06 nM	50.0 – 250 nM	Pharmaceutical formulations Human urine	NR	[75]
	ZIF-67/g-C_3N_4-GCE	DPV	0.071 μM	0.20 – 2.20 μM	Human urine	95.38 – 104.96	[76]
	f-MWCNTs-GO/AuNPs	DPV	0.09 μM	0.4 – 80 μM	Tap Water Tablet Human urine	102.00 98.80 101.00	[43]
	CS/TiO_2/CPE	DPV	0.013 μM	0.2 – 10 μM 10 – 100 μM	Urine Tablets	NR	[77]
	BiOI/GCE	DPV	0.11 μM	0.5 – 11.3 μM 11.3 – 76 μM	River water Human urine	96.50 – 98.50 96.70 – 99.00	[62]
	Oxygen plasma ZnO/SPE	CV	0.11 μM	0.01– 40 μM	Urine	92.00 – 98.00	[78]
	Au/SAM/hydrogel/Apt	EIS	0.02 nM	30 pM – 1 μM	NR	NR	[39]
NAP	GO/GCE	DPV	1.94 μM	10 μM – 1mM	Pharmaceutical formulations	96.90	[52]
	Al_2O_3/GCE	BIA-AMP	0.01 μM	0.5 – 50 μM	NR	NR	[79]
	FTO/$BiVO_4$/CuO	CA	5 nM	5 – 480 nM	Sewage	98.60 – 115.10	[80]
	PPy-CMIP/PE	Potentiometry	0.20 μM	0.4 – 10,000 μM	Serum Saliva	90.70 – 110.20 101.40 – 109.00	[67]
	3D-printed CB/PLA	AMP	0.9 μM	2 – 50 μM	Pharmaceuticals Tap Water	95.00 - 103.00 94.00–102.00	[81]
	Au@f-CNT/GO/GCE	AMP	0.01 μM	0.1 – 15 μM	Water Blood Urine	96.23 – 99.67 97.52 – 99.11 98.53 – 99.19	[38]
	TrGOE, F-GOE, B-rGOE, N-rGOE GO/GCE	DPV	1.94 μM	10 – 1000 μM	Tablet	96.90	[52]
	CeO_2/CuO/MnO, CeO_2/CuO/NiO and La-doped CoFe LDH semiconductor (p-type)	ISFET	11.1 – 19.1 μM	0 – 400 μM	Urine	92.81 – 100.08	[82]
	Poly(L-serine)/GCE PLS/GCE	CV/LSV	0.69 μM	4.3 – 65 μM	Water	101.50 – 106.50	[83]
	Poly(r-oNBA)/GQDs/CPE	DPV	0.672 μM	10 – 100 μM	Tablets	96.70 – 102.00	[84]
	GCE/PEC–BA/AEL	DPV	0.11 μM	25 – 1000 μM	Human urine Human plasma	NR	[85]
	GCE/rGO-TsPro-Cs	DPV	4.1·10 – 7 M	2.0×10^{-5} – 5.0×10^{-4} M	Human urine Human blood plasma	95.30 – 99.20 94.40 – 98.40	[53]
	Gr/$ZnFe_2O_4$/CPE	DPV	0.222 nM	0.05 – 3 μM 3 – 50 μM	Tablet Human serum	97.10 – 100.12 98.25 – 101.14	[68]
MFA	GLU/CPE	SWV	1.01 nM	25 nM – 500 μM	Human urine Tablets	92.70 – 99.16 96.00 – 98.50	[65]
	Cu NCs/CuE	DPV	1 nM	4 nM – 10 mM	Human serum Pharmaceutical formulations	NR	[45]
	$Cu_5V_2O_{10}$/CPE	DPV	2.34 nM	0.01 – 470 μM	Blood serum Pharmaceutical formulations	98.30 – 110.00 93.60 – 106.70	[64]
	$Co_3V_2O_8$/CPE	CV	0.0257 μM	0.090 – 900 μM	Human blood plasma Human urine	99.33 – 102.07 99.14 – 102.80	[19]
	Hf-WO_3/CPE	SWV	3.94 nM	0.01 – 100.0 μM	Urine Tablet	95.50 – 99.30 97.30 – 98.70	[86]

(*continued on next page*)

Table 2 (*continued*)

ANALYTE	MODIFIED ELECTRODE	METHOD	LOD	LINEAR RANGE	REAL SAMPLE	RECOVERY (%)	REF.
	TTN/CPE	DPV	0.0024 μM	0.01 – 400 μM	Wastewater Tablet Human blood serum	93.30 – 110.00 98.00 – 101.00 92.00 – 107.00	[17]
	BDDE	DPV	0.078 μM	0.5 – 100 μM	Pharmaceutical formulations	94.00 – 107.00	[87]
	SFBF-NC/GCE	DPV	2.12 nM	0.01 – 450 μM	Blood Urine Tablet	95.0 – 102.70 96.30 – 101.00 94.60 – 98.40	[18]
	RHG/CPE	SWV	2.13 nM	0.001 – 6000 μM	Human urine	86.50 – 98.20	[66]
	CVO/RGO/GCE	DPV	0.0079 μM	0.001 – 425 μM	Blood serum Human urine	98.50 – 99.40 98.60 – 99.50	[55]
IBU	AgNPs@Af-GO-MIP/ GCE	DPV	0.0087 μM	1 – 100 μM	Human urine Infected human urine Pharmaceutical formulations	97.80 – 100.20 99.42 – 102.20 99.90 – 100.10	[40]
	FULL/CNFPE	DPV	0.180 nM	NR	Surface water	97.60 – 99.10	[11]
	GE	SDLSV	0.6 μM	2 – 500 μM	Tablet	99.20 – 101.60	[88]
	Cu_3TeO_6/GCE	DPV	0.017 μM	0.02 – 5 μM	Human blood serum Human urine	80.00 – 98.60 95.00 – 96.60	[44]
	HNT/PPY/GCE	DPV	0.9 μM	0.1 – 100 μM	Urine	NR	[21]
	NDCNDs/CoTAPhPcNPs/GCE	DPV	4.19 nM	NR	Wastewater	91.20 – 96.30	[89]
ASP	SPCE/(Cs + AuNPs)	CV	0.03 pg/mL	1– 1×10^6 pg/mL	Human urine Saliva	90.00 – 98.00 93.00	[25]
	SEP/MWCNTs/pPGE	DPV	0.047 μM	0.156 – 60 μM	Pharmaceuticals formulations	101.02	[47]
	NDCNDs/CoTAPhPcNPs/GCE	DPV	9.66 nM	NR	Wastewater	100.40 – 101.80	[89]
	CP-BDDE	SWV	0.078 μM	3.96 – 42.75 μM	Pharmaceuticals formulations Urine Tap water	NR	[90]
NIM	NC/CPE	SWV	1.01 nM	0.01 – 0.35 μM	Pharmaceutical formulations Urine	96.00 – 98.60 NR	[91]
	PrW/SPCE	CV	0.007 μM	0.001 – 131.53 μM	Water	96.45 – 99.12	[28]
	$ZnFe_2O_4$ NPs/GCE	DPV	0.006 μM	0.001 – 166 μM	Human urine Tablet	94.60 – 99.60 99.10 – 99.70	[92]
IDM	MCPE	SWV	0.068 μM	0.1 – 100 μM	Pharmaceutical injectable	100.05 – 106.84	[29]
	BDDE	DPV	0.057 μM	0.5 – 100μM	Pharmaceutical formulations	95.00 – 103.20	[87]
	CS-Cu NCs/CuE	DPV	1 nM	1 nM – 10 mM	Human plasma	NR	[45]
PRX	Pt-NiO/MWCNTs/GCE	DPV	0.061 μM	0.6 – 320.0 mM	Human serum urine	98.70 – 102.10 97.40 – 98.30	[48]
	$La(Fe_{0.7}Mn_{0.3})_{0.9}Pd_{0.1}O_3$/GO/GCE	CV	0.89 μM	6 – 103 μM	Human plasma	96.41	[31]
KTP	CPS	DPV	0.11 μM	0.088 – 1.96 μM	Wastewater Horse mackerel	82.30 – 96.40 80.70 – 97.20	[34]

NR: Not reported, DPV: Diferential pulse voltammetry, CV: Cyclic voltammetry, EIS: Electrochemical impedance spectroscopy, LSV: Linear sweep voltammetry, SWV: Square wave voltammetry, SWASV: Square wave anodic stripping voltametric, SDLSV: Semi-Derivative Linear Sweep Voltammetry, AdSDPV: Adsorptive stripping differential pulse voltammograms, CA: chronoamperometry, AMP: Amperometry, BIA-AMP: batch injection analysis – amperometry, BDDE: Boron-doped diamond electrode, GCE: Glassy carbon electrode, SPCE: Screen printed carbon electrode, MIP: Molecularly imprinted polymer, GE: Graphite Electrode PE: Paper Electrode, CPE: Carbon paste electrode, SPCE: screen printed carbon electrode, SPE: Screen-printed electrode, MCPE: modified carbon paste electrode, GPGE: graphenized pencil graphite electrode, CNFPE : carbon nanofiber paste electrode, SPCE: screen-printed carbon electrode, CS-Cu NCs/CuE: chitosan modified copper nanocubics coated on copper electrode, BIA: batch-injection analysis, GO: Graphene Oxide, TrGO: Thermally Reduced Graphene Oxide, B-rGO: Boron-doped partially Reduced Graphene Oxide, F-rGO: Fluorine-doped partially Reduced Graphene Oxide, N-rGO: Nitrogen-doped partially Reduced Graphene Oxide, ISFET: Ion-sensitive field effect transistor, PLS: poly(l-serine), SEP: nano-sepiolite clay, SFBF-NC: Ferrite nanocomposite, RHG: rice husk-derived graphene, CVO: $Ca_{10}V_6O_{25}$, GLU: Glucose, CPS: Carbon paper-based sensor, FULL: Fullerene, GO: graphene oxide, AEL: AlPO4–11 zeolite, SWCNT: Single walled carbon nanotube, AuNPs: gold nanoparticles, f-MWCNTs: functionalised multi-walled carbon nanotubes, ZIF-67/g-C_3N_4: zeolite imidazolate framework-67/graphitic carbon nitride, PEDOT: Polymer 3,4-Ethylenedioxythiophene, CNTs: carbon nanotubes, IONPs: iron oxide nanoparticles, RGO: reduced graphene oxide, PtNFs: platinum nanoflowers, CFYM: chicken feet yellow membrane, LDH: Layered double hydroxid, NC: Nanoclay Evaluating the studies in Table 2, it was seen that electrochemical sensors were tested in aqueous concentration matrices, primarily wastewater and sewage, but also in surface waters (river, tap, lake/sea water). The matrix effect is negatively affected by filtration or signal loss from organic matter, ions, and other cells in the sample. A detailed examination of the studies in Table 2 indicates that matrix effects affect sensor performance in environmental waters. However, the authors emphasize that these results could be addressed with improved pretreatments in the future. Strategies such as sample pretreatment (dilution, filtration) and high-selectivity sensor design (utilizing polymers for segment interconnection) are employed to mitigate these effects.

stability and biocompatibility (Fig. 3). The electrochemical behavior of the sensor was studied using CV and DPV, confirming increased oxidation peak currents due to the synergistic effect of CS and Cu NCs. The proposed sensor exhibited a wide linear range of 1 nM to 10 mM, with a LOD of 1 nM for IDM. Furthermore, the sensor showed excellent repeatability and stability over multiple cycles. Real-time analysis of human plasma samples demonstrated high accuracy and recovery, making the CS-Cu NC/GCE sensor a promising tool for pharmaceutical and biomedical applications.

Monsef et al. [64] introduced an innovative and cost-effective method for the synthesis of vanadium-based nanostructures utilizing a template-mediated approach. The engineered nanostructures,

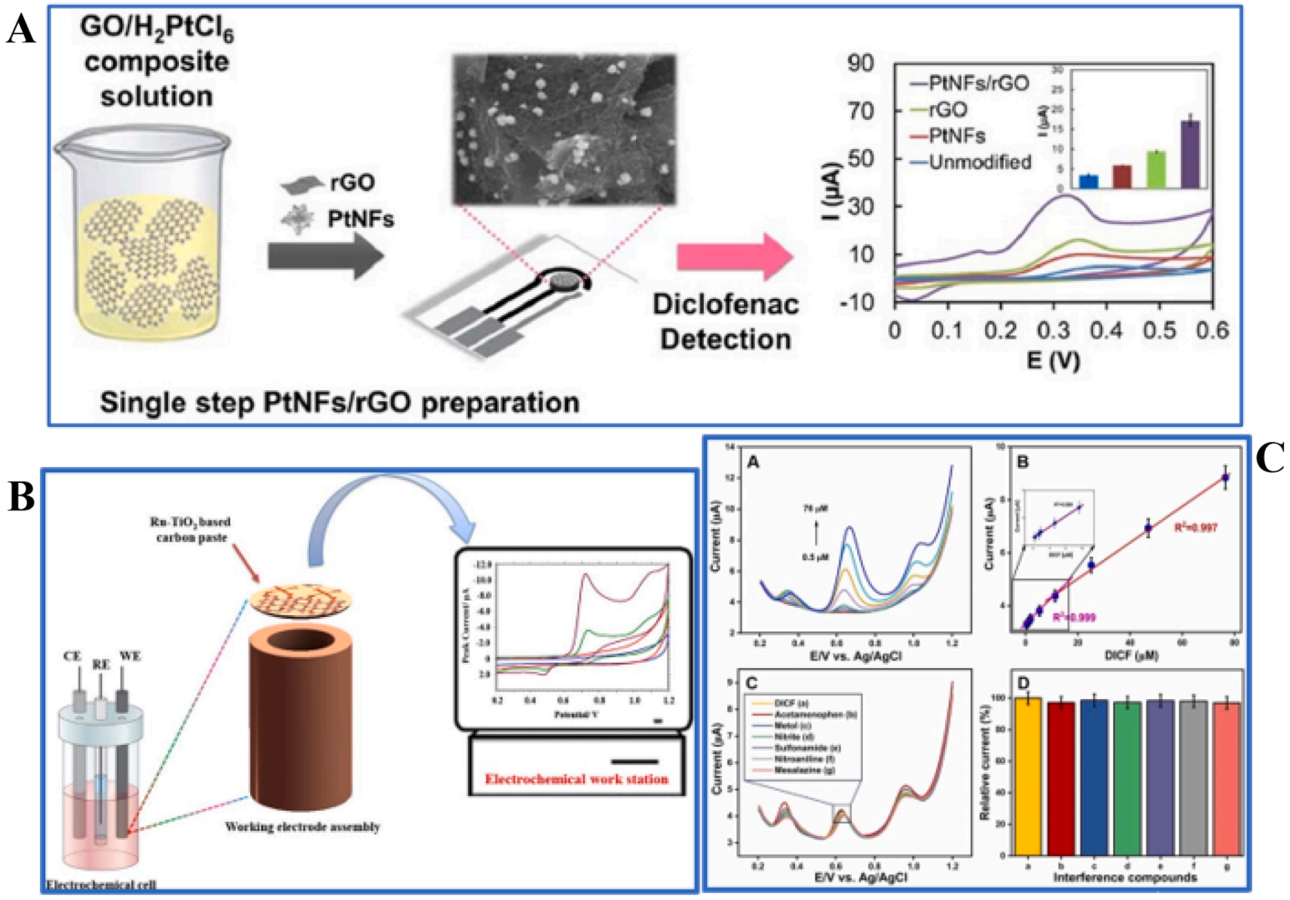
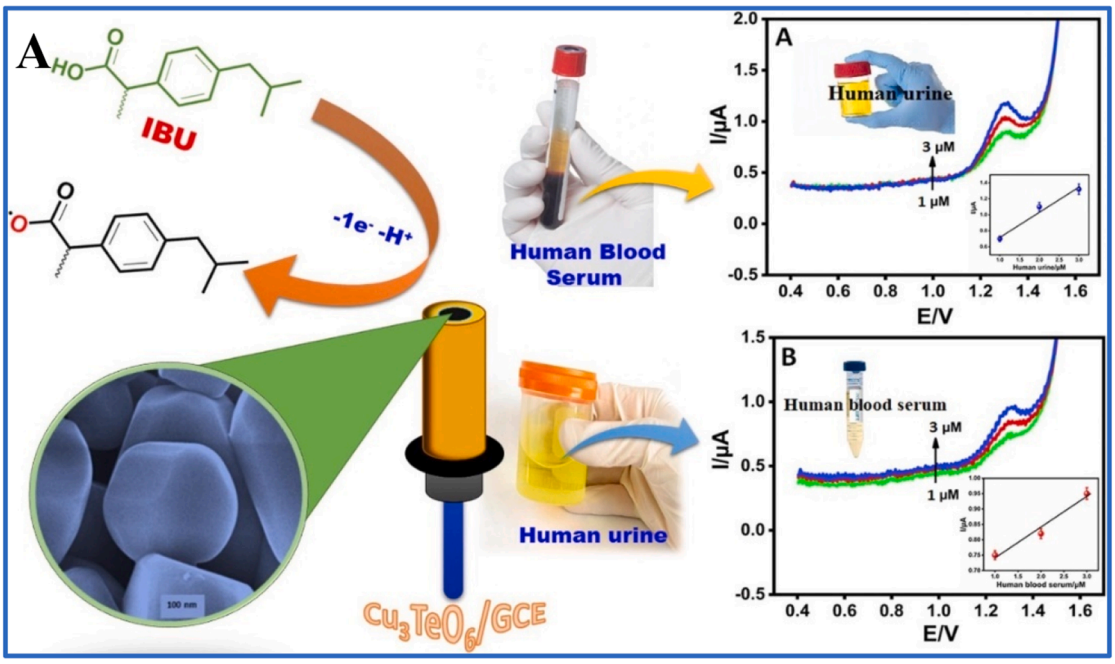


Fig. 1. (A) Design of PtNFs/rGO for sensitive determination of DIC. (B) design of Ru-TiO_2 for sensitive determination of DIC. (C) A) DPV response of different concentrations of DIC in 0.05 M PBS at pH 7.0 on the BiOI/GCE. (B) Dependence of the peak current on the DIC concentration. (C) DPV response of DIC on the BiOI/GCE in the presence of potential interference compounds. (D) corresponding histogram diagram of the interference study. Reprinted from [62,54], and [63] with permission from Elsevier.

characterized by a monoclinic structure and a BET surface area of 18.66 m^2/g, were employed as an electrochemical sensing platform for the ultra-trace detection of methyl farnesoate (MFA). Under optimized conditions of pH 6.0 and a modifier loading of 12.0 %, the electrochemical assays demonstrated a wide linear range of 0.01– 470.00 μM for MFA, with a LOD of 2.34 nM. This work highlights the potential of these nanostructured materials in developing sensitive and robust electrochemical sensors for environmental monitoring applications (Fig. 4**A).** Vernekar P. et al. [65], have developed a new and powerful glucose-modified carbon paste electrode (CPE) for the highly sensitive analysis of MFA. The fabricated sensing probe was characterized using various electrochemical techniques. Electrochemical analysis of MFA was also performed. The sensor's calibration range in the electrochemical analysis was found to be 2.5×10^{-8} to 5.0×10^{-4} M, with a limit of detection (LOD) of 1.01 (±0.03) nM. Experiments with interfering substances were conducted to prove the selectivity of the designed sensor. Real sample analysis was performed on human urine samples and pharmaceutical tablets using SWV, and higher recoveries of 92.70–99.16 % and 96.0–98.5 % were obtained, respectively. This suggests that the sensor can also be used for the detection of MFA in other biological, pharmaceutical, or environmental samples (Fig. 4**B).** In another study, Malode S. et al. [66], colleagues designed a sensor by incorporating graphene nanoparticles produced from RH into a carbon paste to create a unique electrode for the electrochemical analysis of MFA using SWV and CV methods. The RHG/CPE sensor was found to have a detection limit of 2.13 nM and a sensitivity range of 1.0×10^{-8} to 4.0×10^{-4} M. To demonstrate the sensor's selectivity, they tested it with interfering substances found in human urine, blood serum, and breast milk samples, and the results demonstrated its selectivity. The study demonstrates the potential for using RHG/CPE sensors for the selective and sensitive analysis of MFA in biological, pharmaceutical, or environmental samples (Fig. 4**C).**

A simple, low-cost, eco-friendly, highly sensitive, and selective electrochemical sensor for detecting NAP in environmental water and biological samples was developed by Alagumalai et al. [38]. The sensor was based on a hybrid nanocomposite of Au NPs, f-CNT, and GO fabricated via an ultrasound-assisted sonication method. A GCE was used as the substrate for constructing the sensor, demonstrating enhanced electrocatalytic properties. Under optimized conditions, the Au@f-CNT/GO sensor exhibited a wide linear range from 100 nM to 15.6 mM for NAP detection with an LOD value of 14 nM. Furthermore, the sensor demonstrated excellent selectivity, stability, reproducibility, and repeatability. Recovery studies conducted in environmental water (tap, pond, river, and seawater) and biological samples (blood serum and urine) showed recovery rates ranging from 96 % to 99 %. The fabricated Au@f-CNT/GO sensor proved to be a reliable and eco-friendly tool for NAP detection in environmental and biological applications (Fig. 5**A).** Haghgouei H. et al. [67] designed a sensor using an electrochemical synthesis method to modify the surface of conductive flexible papers with a conductive molecularly imprinted polymer. The modified papers were used in dual applications as absorbents and ion-selective electrodes, respectively, in the EC-SPME method and incorporated into a single chip for the extraction and determination of NAP as a model analyte in biological samples. Each compartment was optimized during this process. They achieved remarkable selectivity against NAP in the concentration range of 4.0×10^{-7} to 1.0×10^{-2} mol L^{-1},



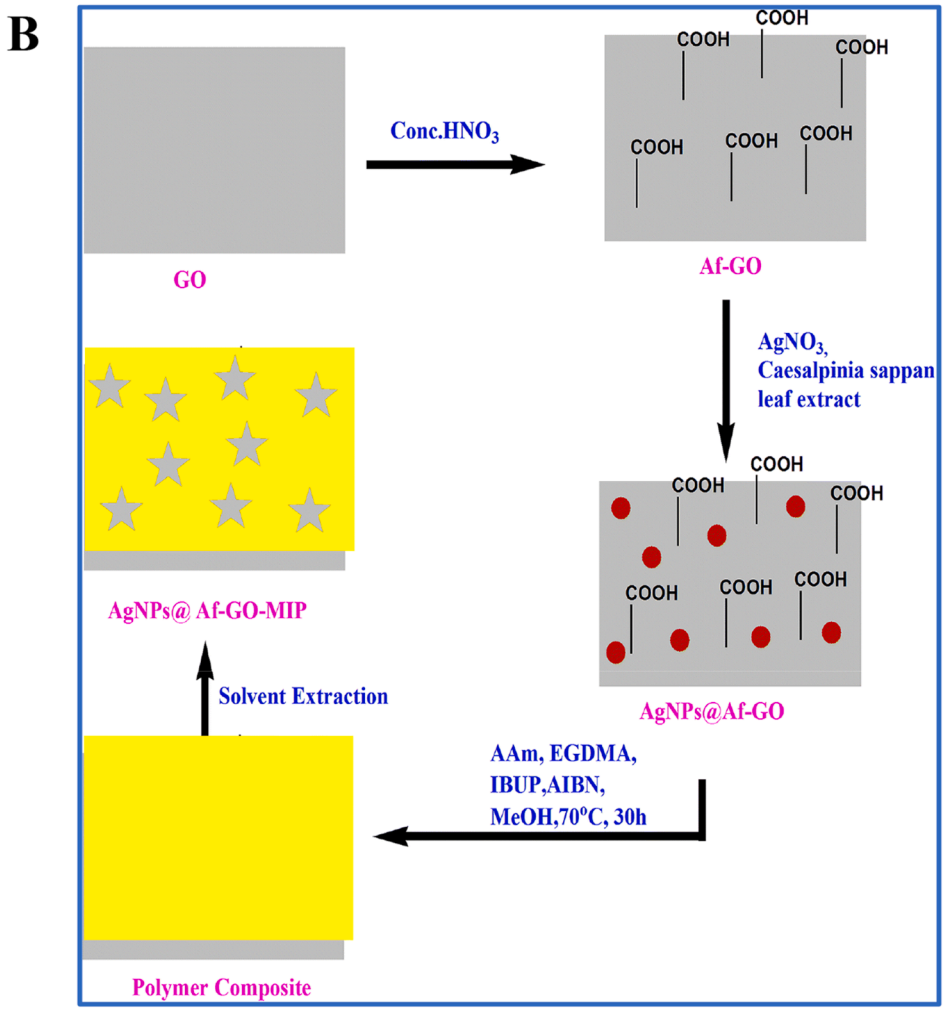


Fig. 2. A) and B) DPV curves for real sample studies of IBU-spiked (0.1 M acetate buffer (pH-5) human urine and human blood serum samples (1–3 μM) using Cu_3TeO_6/GCE at 50 mV/s. (B) design of AgNPs@Af-GO-MIP. Reprinted from [44] and [40] with permission from Elsevier.

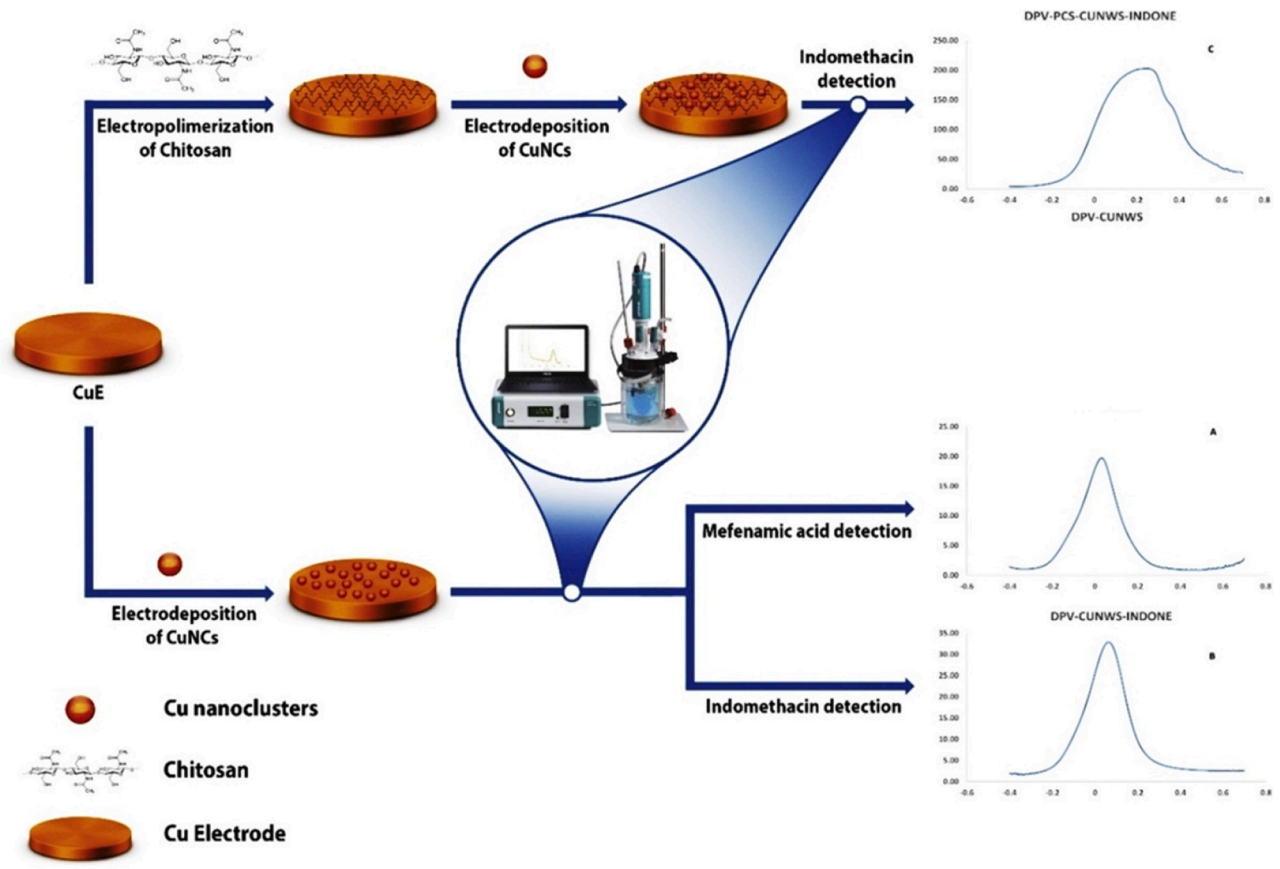


Fig. 3. Schematic representation of sensor fabrication. Reprinted from [45] with permission from Elsevier.

with a coefficient of determination ($R^2 = 0.99$) and a limit of detection (LOD) of 2.0×10^{-7} mol L^{-1}. Satisfactory spike recovery values were found in the acceptable range of 90–110 % with RSDs $\leq$ 7 % for analysis of real samples with a portable LOC device (Fig. 5**B).** Soltani N. et al. [68], designed a sensor with a carbon paste electrode (CPE) modified with graphene and $ZnFe_2O_4$ spinel nanoparticles (Gr/$ZnFe_2O_4$/CPE) for the analysis of NAP. Various electrochemical methods, including cyclic voltammetry (CV), electrochemical impedance spectroscopy (EIS), differential pulse voltammetry (DPV), chronoamperometry, and chronoculometry, were used to study the electrochemical behavior of NAP. The sensor design optimized parameters such as the amount of graphene and $ZnFe_2O_4$ nanoparticles, the pH of the medium, and the device parameters. In the NAP analysis using the DPV method, the concentration was linear in two ranges: 0.05 to 3.00 μM and 3.00 to 50.00 μM, with a detection limit of 0.222 nM. They investigated the interference effects of different species on the selectivity of the designed sensor. The amount of NAP in real samples was measured using a modified electrode. Finally, Gr/$ZnFe_2O_4$/CPE was used to measure NAP in the presence of diclofenac and ascorbic acid (Fig. 5**C).**

Sriram et al. [28] reported a simple and effective CTAB-assisted hydrothermal method to synthesize praseodymium-based metal oxides (PrM) with M site variations (Mo, V, and W). The PrW-modified SPCE nanosensor was applied to the electrochemical determination of NIM using CV, EIS, and amperometric (i-t) techniques (Fig. 6). Under optimized conditions, a linear detection range of 0.001–131.53 μM was achieved for nimesulide, with an ultra-low LOD of 0.007 μM. The PrW/SPCE sensor exhibited excellent selectivity, reproducibility, and operational stability, with minimal interference from co-existing compounds such as glucose and uric acid. Furthermore, real-time analysis of water samples spiked with known concentrations of nimesulide demonstrated recovery rates between 96.45 % and 99.12 %, validating the sensor's practical applicability. The enhanced performance of the PrW-modified SPCE electrode was attributed to its large surface area, abundant active sites, and superior electron transfer kinetics facilitated by the well-defined morphology of PrW nanoparticles. This study highlights the potential of PrW-based electrochemical sensors as a sustainable and efficient tool for environmental monitoring and pharmaceutical analysis.

Finally, in the study by Mohseni et al. [31], a $La(Fe_{0.7}Mn_{0.3})_{0.9}Pd_{0.1}O_3$/graphene oxide (GO) nanocomposite was developed as a perovskite-based electrochemical sensor for the determination of PRX in pharmaceutical and biomedical samples. The electrochemical sensor was fabricated by modifying a GCE with the synthesized $La(Fe_{0.7}Mn_{0.3})_{0.9}Pd_{0.1}O_3$/GO nanocomposite (Fig. 7). The sensor exhibited remarkable analytical characteristics, including a LOD of 0.89 μM and a wide linear range from 6 to 103 μM, achieved through DPV techniques. The integration of GO into the perovskite structure significantly enhanced the sensor's porosity, conductivity, and electrocatalytic activity, leading to an increased electroactive surface area. After optimizing critical parameters such as buffer concentration, electrolyte type, and pH, the sensor demonstrated excellent repeatability, reproducibility, and stability under optimal conditions. The sensor's practical applicability was confirmed by detecting piroxicam in human plasma and pharmaceutical samples with recovery rates of 96.41 %. This innovative $La(Fe_{0.7}Mn_{0.3})_{0.9}Pd_{0.1}O_3$/GO nanocomposite-based electrochemical sensor provides a sensitive, selective, and stable platform for piroxicam determination, paving the way for further advancements in pharmaceutical and biomedical electrochemical applications.

5. Conclusion and future perspectives

NSAIDs are among the most widely consumed pharmaceutical agents due to their broad therapeutic applications and OTC availability. However, their extensive and often uncontrolled usage poses significant

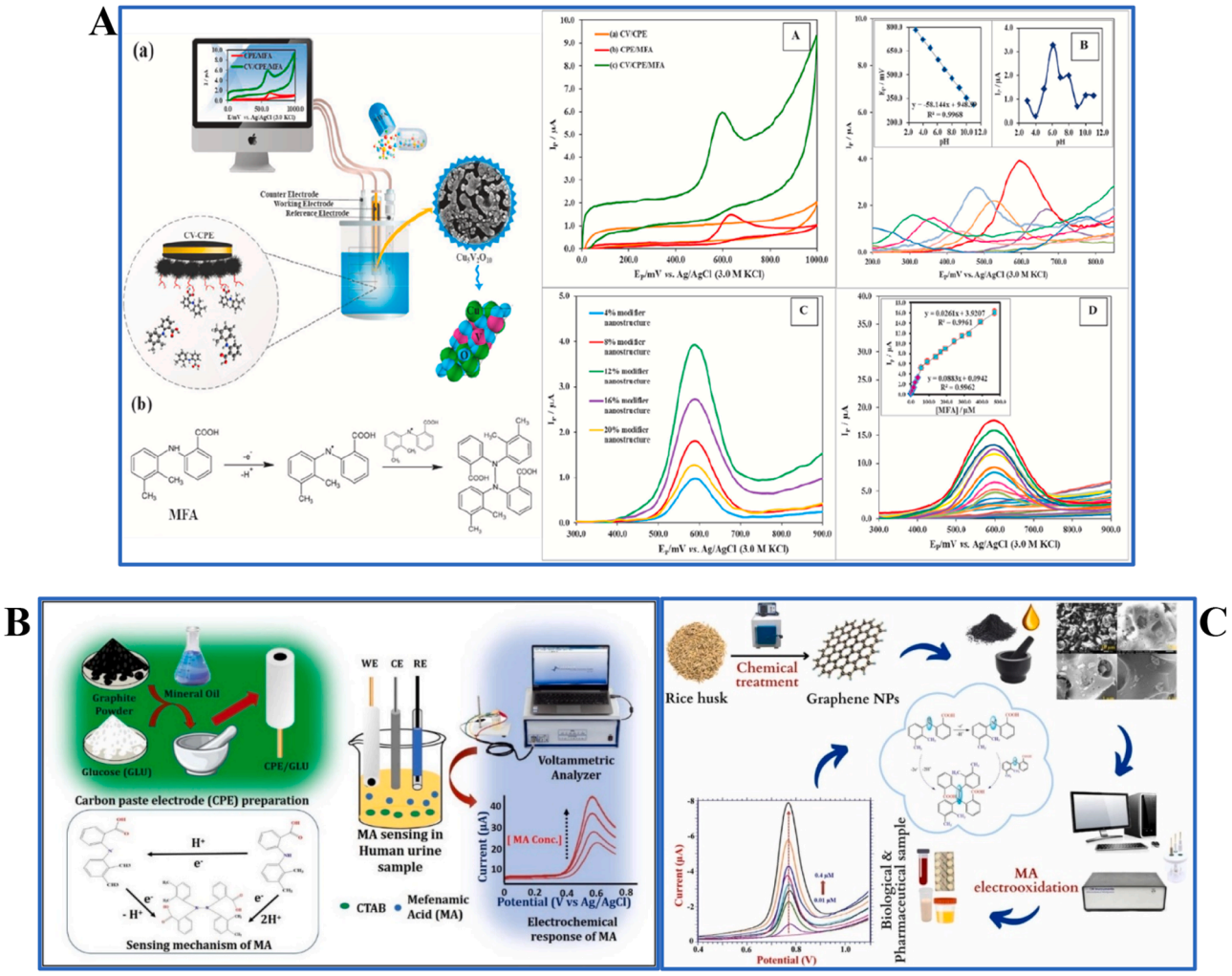


Fig. 4. (A) (a) Schematic diagram, and (b) The catalytic mechanism of MFA at the surface of CV/CPE. And (A) Cyclic voltammograms of 0.2 M BR-buffer (pH = 6.0) at scan rate 100.0 mV s^{-1} for (a) CV/CPE, (b) 40.0 μM MFA at the surface of CPE, (c) as (b) at the surface of CV/CPE, (B) DPVs of MFA at the surface of CV/CPE at various pH of 3.0–11.0, corresponding potential responses (E_P) of MFA vs. pH and I_P of MFA vs. pH at the surface of CV/CPE., (C) effect of different percentage of modifier on the CPE in a medium containing 40.0 μM MFA., and (D) DPVs of the anodic peak response in concentration windows of 0.01–470.00 μM MFA in 0.2 M BR-buffer (pH = 6.0) and corresponding linear calibration plot of the peak response vs. MFA concentration. (B) diagrammatic representation of the fabrication and operating principle of the sensor probe for detecting MFA. (C) design of RHG/CPE sensor. Reprinted from [64,65], and [66] with permission from Elsevier.

health risks and contributes to environmental contamination. The development of sensitive, selective, and cost-effective analytical tools for the determination of NSAIDs in biological, pharmaceutical, and environmental matrices has become increasingly important. Electrochemical sensors have emerged as powerful platforms in this context due to their operational simplicity, rapid response, high sensitivity, and compatibility with miniaturized and point-of-care formats. Since 2020, a growing number of studies have reported the integration of nanomaterials—such as metallic nanoparticles, carbon-based materials, and hybrid composites—into electrochemical sensor platforms to improve performance metrics, including conductivity, surface area, and stability. Among the analytes reviewed, DIC and NAP were most frequently targeted, with DPV and SWV being the most commonly employed techniques, often in conjunction with GCE or SPCE. Urine and tablet dosage forms represented the most frequently analyzed sample types, consistent with clinical and pharmaceutical relevance. The reviewed studies demonstrated detection limits reaching the nano- to picomolar range, underscoring the analytical potential of these platforms for drug quality control, pharmacokinetic monitoring, and environmental surveillance.

In addition to their promising analytical performance, further research is needed to enhance the real-world applicability of these sensors. While many studies report excellent short-term reproducibility and acceptable recovery values in complex matrices, there remains a notable lack of standardized validation protocols. Interference studies, long-term reproducibility assessments, and evaluations under real environmental or physiological conditions are still limited. Future studies should adopt more rigorous and systematic approaches to sensor validation in real samples to ensure accuracy, robustness, and reliability across different settings.

Moreover, although recent efforts have focused on sensor miniaturization and sensitivity optimization, other critical aspects such as regulatory compatibility, field-deployability, and sustainability remain underexplored. The transition from laboratory-scale devices to practical, routine tools will require standardization of testing protocols, assessments of long-term operational stability, and integration with portable or automated systems. Additionally, aligning sensor development with green analytical chemistry principles and regulatory expectations will accelerate adoption in both environmental and biomedical fields.

To further improve analytical throughput and user convenience, future research should also prioritize the development of multi-target sensors capable of detecting multiple NSAIDs simultaneously. Innovations in wearable, smartphone-integrated, and disposable electrochemical platforms hold particular promise for point-of-care and personalized health monitoring applications.

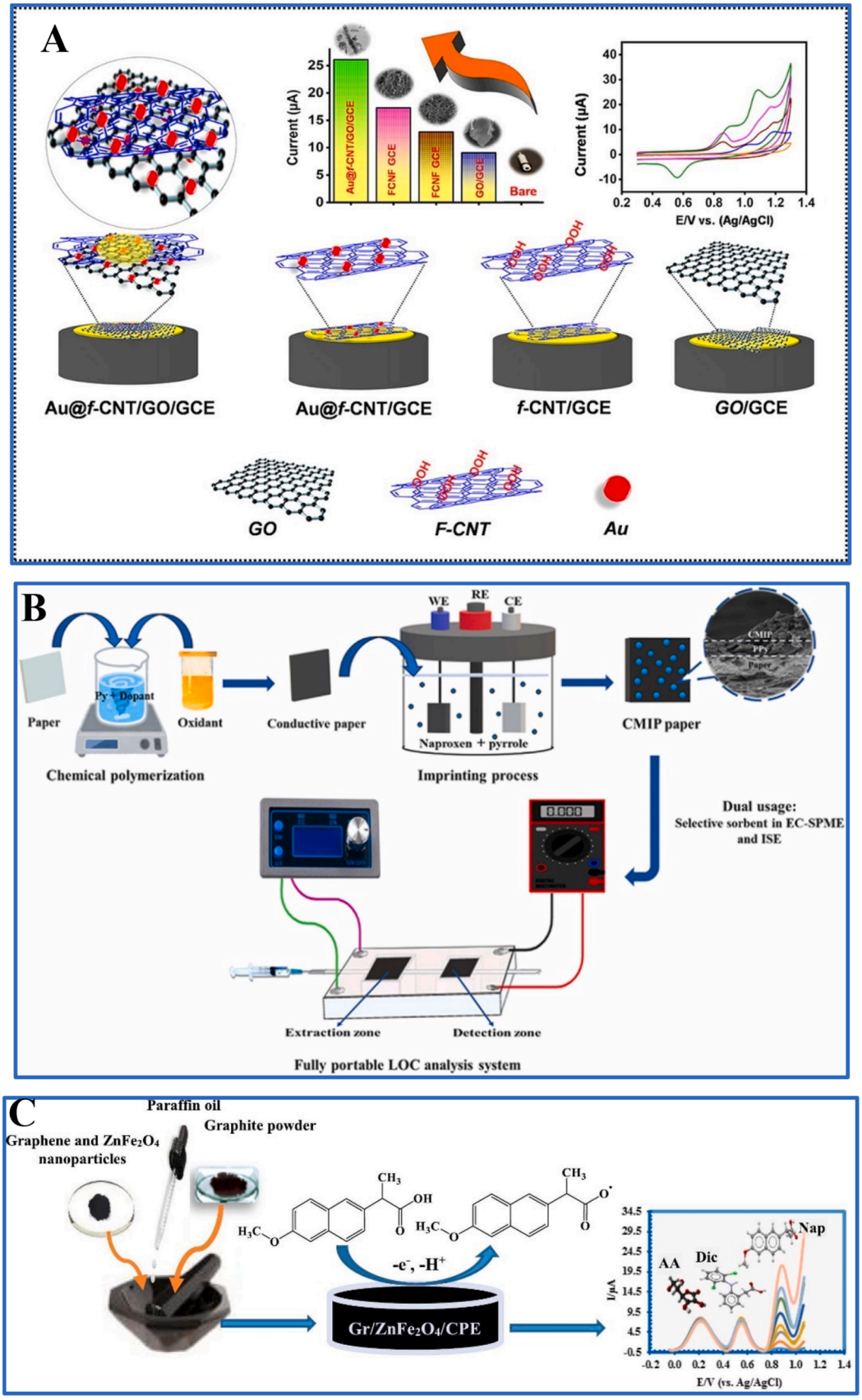


Fig. 5. (A) An electrochemical sensor based on Au@f-CNT/GO/GCE for detection of NAP, (B) conductive papers prepared for NAP analysis by PPy-CMIP were prepared using a chemical procedure, (C) schematic representation of NAP analysis with Gr/ZnFe2O4/CPE sensor. reprinted from [38,67] and [68] with permission from Elsevier.

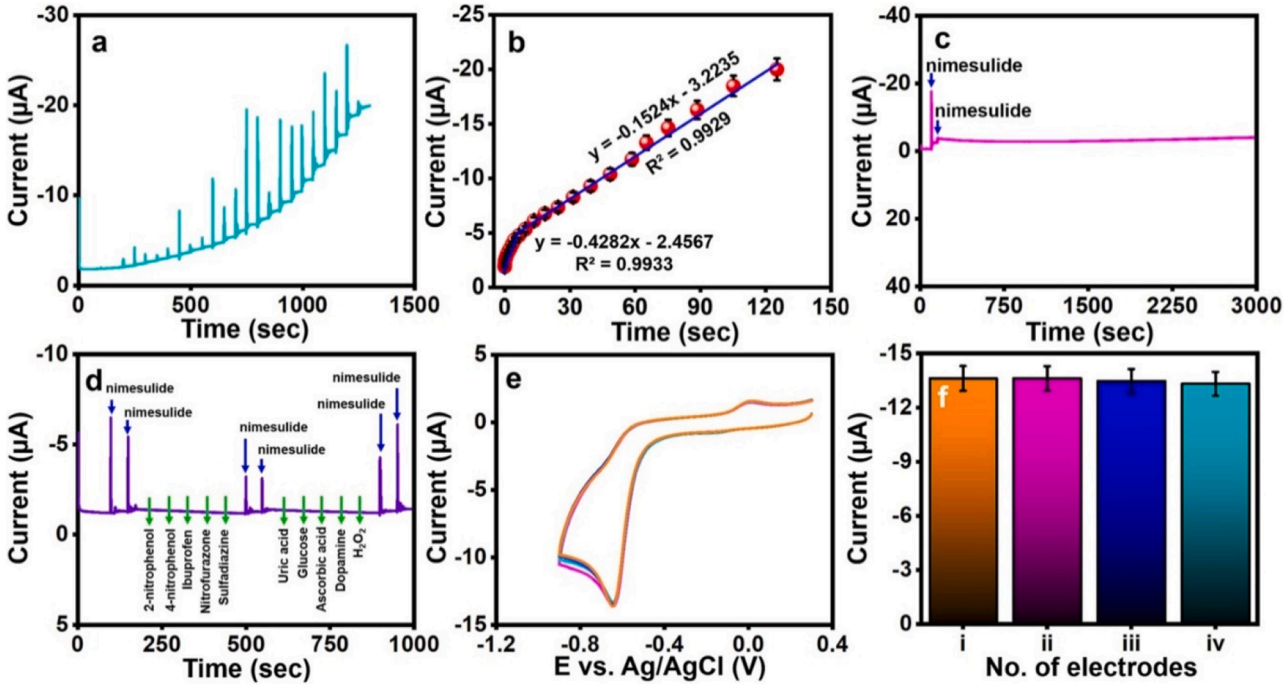


Fig. 6. (a) Amperometry i-t response of various concentrations of nimesulide towards PrW modified electrode with the respective linear plot for concentration versus currents (b). (c) operational stability, (d) selectivity, and (e, f) reproducibility of PrW modified electrode with the presence of nimesulide. Reprinted from [28] with permission from Elsevier.

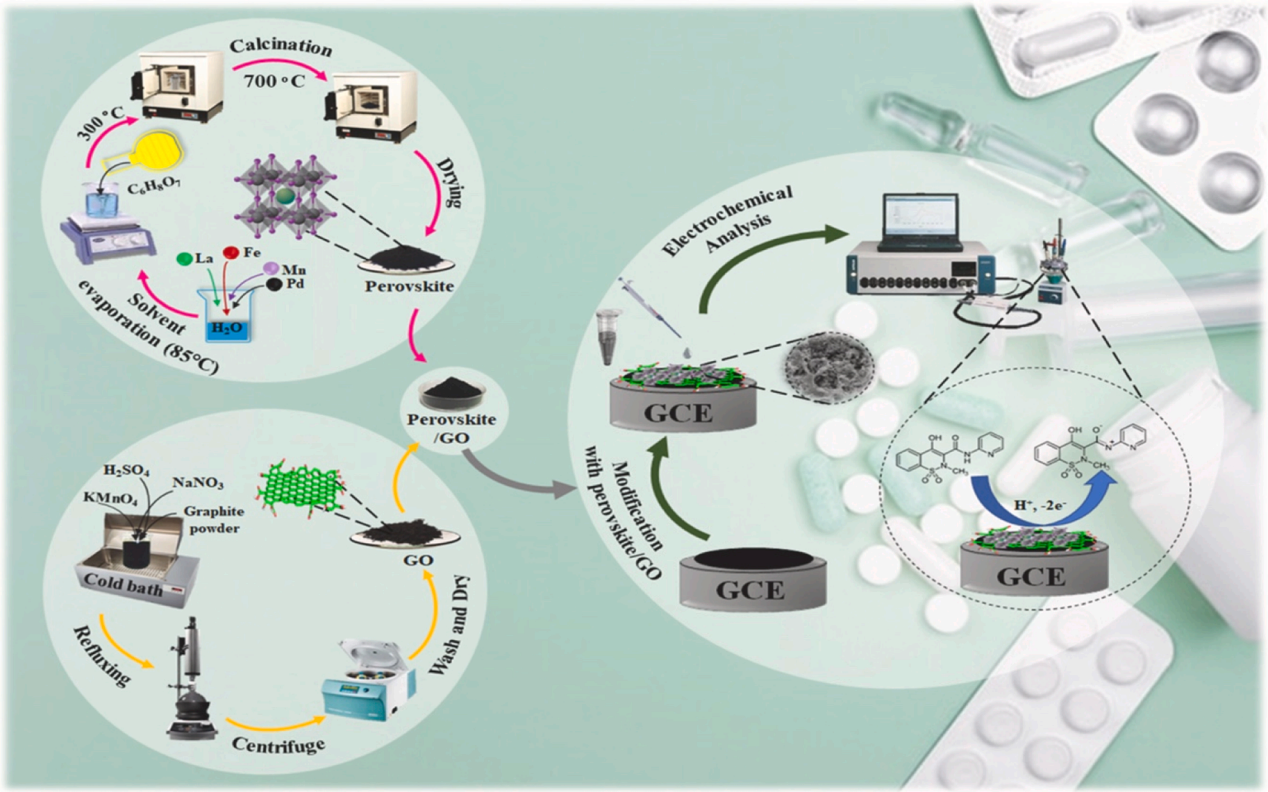


Fig. 7. Graphical representation of synthesis routes regarding nanocomposite fabrication and piroxicam determination. reprinted from [31] with permission from Elsevier.

Looking ahead, the field of electrochemical sensing for NSAIDs is poised to benefit from several innovative directions. The integration of multiplexed sensor arrays will enable simultaneous detection of multiple analytes, significantly improving analytical throughput and efficiency. Advances in data-driven signal processing and artificial intelligence (AI)-based interpretation algorithms could further enhance sensor

precision, especially in complex matrices. Moreover, the convergence of flexible electronics and nanomaterials offers exciting opportunities for the development of wearable or implantable sensor platforms, enabling real-time therapeutic monitoring in clinical and personalized healthcare settings. These emerging trends will play a pivotal role in shaping the next generation of electrochemical sensors toward greater practicality, intelligence, and adaptability.

While this review followed a material-based structure to evaluate trends across analytes, we incorporated NSAID-specific comparisons to highlight compound-level differences. Future reviews may consider a fully analyte-centered framework for further clarity.

CRediT authorship contribution statement

Zeynep Alakus: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Ensar Piskin:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Fatma Budak:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **S. Irem Kaya:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Ahmet Cetinkaya:** Writing – review & editing, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Sibel A. Ozkan:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors also declare that there is no conflict of interests.

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Data availability

Data will be made available on request.

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